

EDUCATION

Ph.D. in Cybersecurity

University of Colorado, Colorado Springs, USA

Expected: Jul 2026

GPA: 3.79/4.00

- Built Bitcoin Core network logger/sensing pipelines and generated datasets collecting multi-layer P2P, routing/network, and consensus data for security and performance research.
- Apply AI/ML to model, predict, and improve network decisions (e.g., peer beneficialness prediction, detect profile spoofing, and privacy-preserving relay/circuit selection).
- Collaborated with interdisciplinary teams on vulnerability identification and security solutions in distributed systems.
- Managed security research projects in distributed networking architectures, with results published in esteemed conferences and journals.
- Presented research findings at domestic and international conferences, and active participation in IEEE and ACM professional community.

Master of Science in Computer Science

University of Malaya, Malaysia

Feb 2018 – Sep 2020

Bachelor of Science (Engineering) in Computer Science and Engineering

Shahjalal University of Science and Technology, Bangladesh

Jan 2008 – Jun 2012

TECHNICAL SKILLS & COMPETENCIES

Permissionless Blockchains: Studied and modified Bitcoin and Ethereum protocols, enhancing implementations by proposing security mechanisms.

- Performed network sensing and traffic analysis, generated experimental datasets, identified network threats and proposed mitigation strategies.

AI/ML: Applied machine learning techniques for modeling, classification, and prediction in security-focused networks to expose patterns and anomalies.

- Developed predictive models for network behavior and applied classification methods for adversary recognition via network fingerprinting.

Cryptography: Expertise in traditional applied cryptography and post-quantum techniques, with a focus on protocol design and systems analysis.

- Mentored students in cryptography courses emphasize practical usage and implications of the cryptographic principles.

Recommendation Engines: Created intelligent engines capable of generating personalized and context-aware results given user constraints.

- Integrated content-based filtering, collaborative filtering, and data mining techniques to build high-performing personalized models.

Tools and Platforms: Git, Docker, GNU Debugger (gdb), Bitcoin Core, Geth (Go-Ethereum), Wireshark, Scapy, Tcpdump, Nmap, Netcat, Metasploit, OpenSSL, Jupyter, Pandas, NumPy, SciPy, Matlab, Matplotlib, scikit-learn, TensorFlow, PyTorch

Languages: C/C++, Python, Java, Unix/Bash, JavaScript, C#/Visual Basic, Assembly, Ruby, PHP, and Regular Expression pattern matching

Relevant Coursework: Applied Cryptography, Fundamental of Computer/Network Security, Homeland Security & Cyber Security, Computer Communication, Privacy and Censorship, Reinforcement Learning, Networks, Crowds, and Markets

EXPERIENCE

Graduate Teaching Assistant

@ Computer Science Department, University of Colorado, Colorado Springs, USA

Jan 2026 – May 2026

20 hrs/wk

- Instructor for **CS 2080: UNIX Programming**. Taught UNIX programming; prepared and delivered lectures, live-coding demos, and in-class exercises; developed assignments and quizzes aligned with learning outcomes.
- Teaching support for **CS 5300 Advanced Software Engineering**. Led tutoring sessions, graded assignments and exams, and held office hours.
- Teaching support for **CS 2060: Programming with C**. Led tutoring sessions, graded programming assignments and exams, and held office hours.

Graduate Teaching Assistant

@ Computer Science Department, University of Colorado, Colorado Springs, USA

Aug 2025 – Dec 2025

20 hrs/wk

- Instructor for **CS 1150: Principles of Computer Science**. Taught Java programming; prepared and delivered lectures, live-coding demos, and in-class exercises; developed assignments and quizzes aligned with learning outcomes.
- Teaching support for **CS 2060: Programming with C**. Led tutoring sessions, graded programming assignments and exams, and held office hours.

Summer Conference Assistant

@ University of Colorado, Colorado Springs, USA

Jun 2025 – July 2025

40 hrs/wk

- Supported youth camps and large-scale conferences by managing operations and setup.
- Collaborated with a diverse team, strengthened skills in communication, problem-solving, and teamwork.

Graduate Teaching Assistantships

@ Computer Science Department, University of Colorado, Colorado Springs, USA

Jan 2025 – May 2025

20 hrs/wk

- Teaching support for **CS 3020: Introduction to C#**. Responsibilities included grading, tutoring, recitation, office hours, and attending classes.
- Teaching support for **CS 3910: System Administration and Security**. Graded, tutored, recitation, attended classes, and held office hours.
- Teaching support for **CS 4920: Applied Cryptography**. Responsibilities included grading, tutoring, recitation, office hours, and attending classes.

Mentored Doctoral Fellowship

@ Graduate School, University of Colorado, Colorado Springs, USA

Aug 2024 – May 2025

20 hrs/wk

- Problem-Solving:** Built an intelligence-enabled low-latency Tor prototype (Quick Tor) for permissionless blockchains.
- Data Analysis:** Measured relay-to-relay and client-to-relay latency, filtered out relays exceeding a threshold (τ), and constructed circuits from the remaining low-latency relays.

Graduate Research Assistant

@ Networked System Security Lab, University of Colorado, Colorado Springs, USA

Jun 2023 – Dec 2024

20 hrs/wk

- Problem-Solving:** Collaborated on strengthening distributed network security by identifying vulnerabilities in permissionless blockchains and designing effective countermeasures against emerging threats.
- Data Analysis:** Investigated P2P protocol behavior and applied machine learning techniques to predict behavior and detect anomalies.

Departmental Teaching Assistant

@ Computer Science Department, University of Colorado, Colorado Springs, USA

Jan 2023 – May 2023

20 hrs/wk

- Teaching assistant for **CS 1450 Data Structures**. Responsibilities included grading, tutoring, recitation, office hours, and attending classes.

Assistant Professor

@ Department of Computer Science and Engineering, Dhaka International University, Bangladesh

Jan 2022 – Dec 2022

40 hrs/wk

▪ Teaching and Curriculum Development: Prepared and delivered lectures, tutorials, and lab sessions for undergraduate and graduate courses. ▪ Developed course materials, syllabus, and assessments to align with academic standard. Supervised projects, internships, and thesis work. ▪ Student Mentorship and Advising: Advise students on academic and career matters. Guide students in research projects, competitions, and extracurricular academic activities.	
Graduate Research Assistant	Feb 2018 — Jan 2020
@ Software Engineering Lab, University of Malaya, Malaysia	20 hrs/wk
▪ Problem-Solving: Developed a recommendation engine leveraging Information Retrieval, Recommender Systems, and Data Mining techniques. ▪ Data Analysis: Built a Recommender System pipeline to crawl and preprocess scholarly data, generated citation-matrix/graph datasets, and applied similarity-based ranking for evaluation.	
Research Fellow, Bangabandhu Science and Technology Fellowship Trust	Feb 2018 — Jan 2020
@ Ministry of Science and Technology, People's Republic of Bangladesh.	40 hrs/wk
▪ Conducted advanced postgraduate research in computer science, focusing on intelligent systems for scientific paper recommendation. ▪ Contributed to the advancement of AI-driven solutions aligned with national priorities in education, research, and innovation.	
Lecturer	May 2014 — Dec 2017
@ Department of Computer Science and Engineering, Dhaka International University, Bangladesh	40 hrs/wk
▪ Teaching and Curriculum Development: Delivered lectures, tutorials, and lab sessions for students; developed and updated course syllabus, assessments, and instructional materials in alignment with academic standards. Supervised student thesis work, projects, and internships. ▪ Student Mentorship and Advising: Provided academic and career guidance to students; mentored student research projects, competitions, and extracurricular activities.	
Junior Software Engineer	Oct 2013 — Apr 2014
@ MicroMac Techno Valley Ltd, Bangladesh	40 hrs/wk
▪ Built dynamic, database-driven web applications using PHP (CodeIgniter) and MySQL, delivering responsive UI. ▪ Contributed to design and development of software solutions for banking clients, focusing on secure data handling and transaction processing. ▪ Collaborated with cross-functional teams to gather requirements, implement features, and ensure software quality through testing/debugging.	

OPEN-SOURCE CONTRIBUTIONS & NOTABLE PROJECTS

Bitcoin Core: Protocol Implementation & Logging Enhancements	Jan 2023 — Present
https://github.com/nsakibuuccs/Bitcoin-P2P-Enhanced-Network-Logging	
▪ Built permissionless blockchains' network logger/sensing pipelines and generated datasets collecting multi-layer P2P, routing/network, and consensus data for security and performance research. ▪ Introduced specialized RPC commands for enhanced system interactions.	
ML-Based Prediction and Classification in Bitcoin	Aug 2024 — Present
https://github.com/nsakibuuccs/Machine-Learning-Prediction-and-Classification	
▪ Integrated predictive and classification pipelines to anticipate peer behavior and detect profile spoofing. ▪ Established sensing, labeling, and training workflows based on live network observations. ▪ Evaluated deployable models based on measured accuracy, stability, and runtime efficiency.	
Cryptography Utility for File Encryption and Signing	Jan 2023 — May 2023
https://github.com/nsakibuuccs/Cryptographic-Primitives-Toolkit	
▪ Built a cryptographic library for RSA-2048 keypair generation, AES-CBC file encryption/decryption, and RSA-PSS signing/verification. ▪ Provided immediate authenticity checks with signature verification feedback to confirm validity or flag tampering.	
Context-Aware Personalized Recommendation Framework	Feb 2018 — Jan 2020
▪ Built an intelligent data pipeline to crawl, preprocess, and structure scholarly metadata, generating citation-matrix and citation-graph datasets. ▪ Applied Information Retrieval, Recommender Systems, and Data Mining techniques.	

AWARDS & HONORS

▪ Recipient: Mentored Doctoral Fellowship, UCCS (Highly competitive award granted to only 4 doctoral students)	Aug 2024 — May 2025
▪ Best Paper Award at 16 th IEEE International Colloquium on Signal Processing & Its Applications (CSPA), Malaysia	Received — Feb 2020
▪ Research Fellow , Bangabandhu Science and Technology Fellowship Trust, People's Republic of Bangladesh	Feb 2018 — Jan 2020

INVITED TALKS & PRESENTATIONS

@ University of Colorado, Colorado Springs, USA	Apr 2025
▪ Lecture: Blockchain and Cryptocurrency, Teaching Assistant- CS 4920 Applied Cryptography	
@ IEEE 21 st Consumer Communications & Networking Conference (CCNC)	Jan 2024
▪ Conference Presentation: Positive Reputation Score for Bitcoin P2P Network	
@ N3SP Spring Forum, US Air Force Academy	Apr 2023
▪ Poster Presentation: Behavior-based Positive Reputation Score for Bitcoin P2P Network	
@ 16 th IEEE International Colloquium on Signal Processing & Its Applications (CSPA)	Feb 2020
▪ Conference Presentation: Renal Cell Cancer Nuclei Segmentation from Histopathology Image Using Synthetic Data	

RESEARCH PUBLICATIONS

Ongoing: 2, Accepted: 1, Published: 11

▪ "DyPBP: Dynamic Peer Beneficialness Prediction for Cryptocurrency P2P Networking"	arXiv Preprint 2026
▪ "Anonymous Networking Detection in Cryptocurrency Using Network Fingerprinting and Machine Learning"	MDPI Electronics 2025
▪ "Analyzing and Modeling Connection Impact on Distributed Consensus in Cryptocurrency Blockchain"	IEEE/IFIP NOMS 2025*
▪ "Network Fingerprinting Using Machine Learning for Anonymous Networking Detection in Cryptocurrency"	IEEE CCNC 2025*
▪ "From Slow Propagation to Partition: Analyzing Bitcoin Over Anonymous Routing"	IEEE ICBC 2024*
▪ "Positive Reputation Score for Bitcoin P2P Network"	IEEE CCNC 2024*
▪ "A Hybrid Personalized Scientific Paper Recommendation Approach Integrating Public Contextual Metadata"	IEEE Access 2021
▪ "A Collaborative Approach Toward Scientific Paper Recommendation Using Citation Context"	IEEE Access 2020
▪ "Renal Cell Cancer Nuclei Segmentation from Histopathology Image Using Synthetic Data" (Best Paper Award)	IEEE CSPA 2020

* Indicates conferences recognized for their selectivity (acceptance rate below 30%).